

Security Consciousness and Internet Service Provider Selection: An Empirical Analysis of Dutch ISP Markets

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Abstract

Market-based approaches assume that informed customer choice can raise cybersecurity standards among Internet Service Providers (ISPs). However, the link between server operators' security consciousness, their ISP selection, and observable network-level security outcomes remains underexplored. This study examines whether server operators with stronger security awareness preferentially choose ISPs that position themselves as security-focused in the Dutch ISP market.

Using Shodan measurements for five Dutch ISPs representing different market positions, a binary indicator of operator security awareness was constructed based on exposure to three well-known, patchable vulnerabilities. Vulnerability rates were compared across mainstream providers (KPN, Ziggo, Odido), a premium provider (Delta Fiber), and a provider that explicitly markets security and privacy as core differentiators (Freedom Internet).

The results disprove the hypothesis. Freedom Internet, which emphasises security and privacy in its marketing, exhibited the highest vulnerability rate (15.11%), significantly exceeding all mainstream providers (9.48–12.73%; $p < 0.025$). This finding directly contradicts the hypothesis that security-conscious operators gravitate toward ISPs that promote security in their market positioning.

The evidence indicates a market failure in cybersecurity governance. Security marketing does not correlate with measurable security outcomes, and customers cannot reliably distinguish provider quality through market signals alone. Voluntary market differentiation appears insufficient to improve collective cybersecurity, suggesting that regulatory intervention establishing minimum security standards and clarifying ISP-operator responsibilities may be necessary to achieve meaningful security improvements.

1 Introduction

In February 2021, a cyberattacker breached a Florida water treatment facility and attempted to poison the water supply by remotely manipulating chemical levels [9]. Investigators traced the vulnerability to an exposed TeamViewer connection [9] discoverable via internet scanning platforms.

Internet scanning platforms, such as Shodan and Censys, have reshaped cybersecurity by reducing reconnaissance from a time-intensive, expert task to near-instant queries accessible to non-experts. As of 2025, they collectively index more than 800 million internet-facing services worldwide [7]. Studies show they cover 98% of services on the ten most common ports [7], supporting defensive assessments while also enabling attackers to find vulnerable

systems at scale. This transparency has a dual effect: it strengthens defensive capabilities yet lowers barriers for offensive targeting.

This ecosystem involves several actors whose decisions jointly shape how vulnerabilities are distributed. Server operators are responsible for securing systems, but often lack clear visibility into their internet exposure. Internet Service Providers (ISPs) supply the network infrastructure through which servers connect to the Internet. ISPs vary widely in their security practices, monitoring capabilities, and customer support. Variations include whether providers actively monitor for compromised hosts, notify customers of detected vulnerabilities, implement filtering to prevent malicious traffic, or provide technical assistance for securing internet-facing services. Regulators must protect critical systems while preserving market flexibility. Understanding how these actors interact, especially the link between operator security awareness and ISP selection, has direct policy implications.

A fundamental tension exists between market-driven security improvements and regulatory intervention. Suppose more security-conscious operators preferentially choose ISPs that market themselves as security-focused. In that case, market forces may sort customers by security posture across providers, suggesting that differentiation can drive improvement without regulation. Conversely, if vulnerabilities are distributed randomly across ISPs regardless of their security marketing, the market is failing, and regulatory action may be warranted. This study examines the Dutch ISP market to empirically test this market mechanism, where diverse ISP options and readily available Shodan data enable a systematic analysis of operator-ISP patterns.

The paper proceeds as follows. Section 2 reviews prior work on internet scanning platforms, ISP security practices and responsibilities, and operator security behaviour. Section 3 states the research question and hypothesis. Section 4 describes the methodology for measuring security consciousness and analysing ISP selection. Section 5 presents the empirical findings. Section 6 discusses limitations. Section 7 concludes with implications for cybersecurity policy and multi-stakeholder governance.

2 Literature

Three distinct yet complementary perspectives have emerged in prior work. The first concerns the technical dimension, focusing on how platforms collect, classify and expose data as well as examining their accuracy and reliability. The second perspective addresses the economic and market position of ISPs, how they hold responsibilities and influence user trust. Thirdly, research focuses on behavioural dimensions, focusing on the security behaviour of individual users. This literature section highlights the perspectives to underscore their respective importance and to identify research gaps in each.

The first research perspective focuses on the technical aspects of internet scanning platforms, in particular Shodan. Research by Tundis et al. investigates how the visibility created by internet scanning platforms influences attack behaviour. They find that devices labelled by Shodan receive significantly more attack traffic than comparable systems that are not listed, suggesting that exposure on such platforms translates to an increased likelihood of being targeted [24]. Building on this, Huang et al. critically evaluates the reliability of commercial internet scanning and vulnerability tagging platforms, showing that while these scanners provide large-scale visibility, their detections frequently show false positives and incomplete coverage [11]. The research emphasises that the data should be interpreted as indicators of potential exposure rather than definitive evidence of a vulnerability. In addition, Bodenheimer et al. demonstrates similar limitations in a focused study on industrial control systems, where Shodan often misqualified exposed devices [2]. Collectively, these studies indicate that Shodan's data offers valuable visibility but must be interpreted cautiously.

The second research perspective examines the role of ISPs in responsibilities for safeguarding networks and the ways in which their security practices and marketing shape user trust and perception. Usman's research evaluates the responsibilities of ISPs towards customer network security [25]. He argues that ISPs occupy a privileged position to detect and prevent malicious activity, given their infrastructure, expertise and visibility of network traffic. However, despite this capability, there is no legal obligation to do this, leaving security largely dependent on voluntary best practices. In contrast, Ray et al. examine how service providers influence perceived security through trust and signalling [20]. Their study shows that users cannot directly evaluate the real security of online providers and instead rely on indirect cues such as branding and communication, meaning that security marketed ISPs may attract trust regardless of their actual performance. These studies demonstrate that ISPs play a critical role, yet it is uncertain whether they deliver on their security claims, as users trust is driven more by perception than by verified performance.

From a human-centric standpoint, research highlights how the persistence of vulnerabilities is also rooted in human factors. Edu and Papadaki investigated the extent to which users and network administrators understand and act upon vulnerability information [8]. Their study showed that formal education or professional experience does not guarantee proactive vulnerability management, underscoring that awareness alone does not translate to action. Expanding this perspective, Mejias et al. developed a model of

information-security vulnerability awareness that integrated individual, organisational and contextual aspects [13]. They argue that factors such as technical competence, risk perception and culture collectively shape whether awareness actually leads to patching their vulnerabilities.

Drawing on this research, gaps remain between these perspectives. Technical studies measure vulnerability exposure but do not connect it to user or provider behaviour. ISP research highlights claims of security without verifying their actual performance. Behavioural studies show that awareness does not necessarily lead to action, but instead leaves a gap in understanding how such awareness influences broader choices, such as ISP selection. This study aims to bridge these gaps by linking behavioural, technical and market dimensions, examining whether operator security consciousness affects ISP choice and translates into measurable differences in vulnerability exposure.

3 Research Question

The introduction highlighted how internet-wide scanning platforms have reshaped the cybersecurity landscape, exposing a tension between market-based security improvements and regulatory intervention. The relationship between operator decision-making and network-level outcomes remains underexplored, creating a critical gap in understanding whether economic incentives improve collective cybersecurity.

Therefore, this study investigates the following research question:

How do different levels of security consciousness among server operators correlate with their Internet Service Provider selection in the Netherlands?

The objective of this study is to empirically test whether market behaviour by server operators produces measurable ISP-level differences in vulnerability exposure. By linking indicators of operator security consciousness to the networks connecting their services, the study examines whether competitive market forces reward providers that market themselves as having stronger security practices, or whether such differentiation fails to occur, indicating a need for policy or regulatory intervention.

3.1 System Architecture

Figure 1 presents the system architecture underlying this study, illustrating how the three key actor groups interact through technical infrastructure to produce observable security outcomes.

The architecture reveals how security decisions are propagated throughout the system. Server operators choose an ISP and deploy internet-facing services via their infrastructure. This choice determines which network their services become visible through. Internet Service Providers supply the network infrastructure that connects these services to the internet, assigning IP address ranges that identify which operator belongs to which provider. Scanning platforms like Shodan continuously probe these IP addresses, detecting services and identifying vulnerabilities in their configurations.

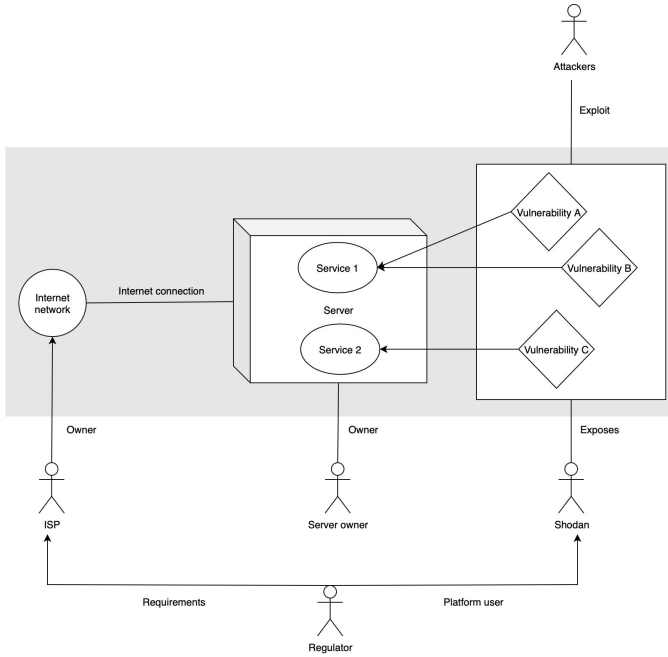


Figure 1: Diagram of the system architecture and the actors' relation to its components.

3.2 Hypothesis

This study formulates the following hypotheses to empirically test the relationship between operator security consciousness and ISP-level vulnerability exposure:

- **H₀:** There is no significant relationship between operator security consciousness and ISP selection. Vulnerabilities are randomly distributed across ISPs, independent of operators' security behaviour.
- **H₁:** There is a significant relationship between operator security consciousness and ISP selection. More security-conscious operators preferentially select ISPs that market themselves as having stronger security practices, leading to lower observed vulnerability rates within those networks.

3.3 Conceptual model

Figure 2 presents the conceptual model linking operator security consciousness, ISP selection, and the distribution of vulnerabilities across Dutch ISPs. It connects individual decisions to aggregated network-level security outcomes.

Security consciousness refers to operators' awareness, knowledge, and proactive actions to maintain secure configurations and remediate known vulnerabilities. Operators with higher security consciousness are expected to support systems with fewer unpatched exposures, as reflected in data from platforms such as Shodan.

ISP choice is treated as a mediating factor. The study evaluates whether operators select ISPs not only based on technical quality or price, but also on how providers position themselves in terms of security. If market-driven sorting occurs, ISPs that promote robust

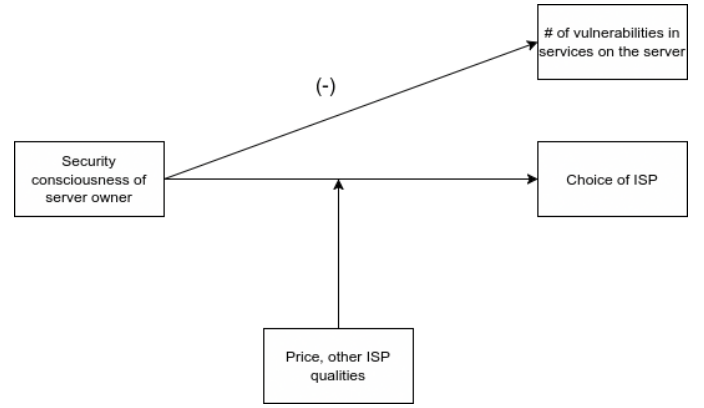


Figure 2: Conceptual model

security standards should attract more security-aware customers and, in turn, exhibit fewer vulnerable hosts in aggregate scanning data.

This relationship may be moderated by other factors, such as price, performance, and availability, which often influence purchasing decisions. In such cases, security-focused marketing may have limited behavioural impact, producing no systematic differences in vulnerability exposure across ISPs and suggesting that the market does not organically reward stronger security practices.

The conceptual model, therefore, contrasts two mechanisms: (1) Security-conscious operators preferentially select ISPs that market themselves as secure, producing measurable differences in ISP-level vulnerability rates. (2) ISP selection is driven primarily by price and performance, yielding no systematic differences in vulnerability exposure and implying a lack of market-based differentiation.

By testing whether vulnerability exposure aligns with ISP market positioning, this study assesses whether observable patterns are consistent with market-driven security differentiation or, instead, indicate the absence of such mechanisms. The findings will inform whether market dynamics can improve cybersecurity outcomes, or whether more direct regulatory measures are warranted.

4 Methodology

4.1 Dataset

Shodan is an IP address search engine that scans the internet for machines that host internet services/protocols, which are referred to as hosts or servers. For each IP address, Shodan tests a wide range of TCP/UDP ports to see if they are open or not. Shodan then attempts to perform a handshake for the most common protocol hosted on a given port in order to try and obtain the banner for the protocol. The open ports and service banners are saved and made accessible to the public via the search engine.

What sets Shodan apart from other IP search engines is that it also tests each service on a host machine for common vulnerabilities. This is done by analysing the service banners, extracting version information and matching this to a database of CVEs (Common Vulnerabilities and Exposures) [23]. Using this simple technique, Shodan is able to annotate vulnerable hosts with corresponding CVE IDs.

One important thing to keep in mind is that honeypots are also present in the Shodan dataset. Honeypots are devices on the internet that mock real services to trick attackers into interacting with them. Data on these interactions collected by honeypots serve as valuable cyber threat intelligence. However, for this research, honeypots are false positives in the data that can skew the results. Therefore, any obvious honeypots are removed from the data. These are hosts that were identified by Shodan as honeypots and hosts that have more than 100 open ports [3].

4.2 Common vulnerability indicator

The data provided by Shodan can be used to create an indicator for the security consciousness of the operators of the servers. Security consciousness in this context can be broadly defined as: the awareness of IT operators that hosting services on the internet comes with certain security risks, the understanding of the importance of addressing weaknesses in the security of the hosted services, and the knowledge to adequately fix known vulnerabilities. On top of that, a security conscious server operator always takes action to fix serious vulnerabilities. In general, operators that have a higher level of security consciousness will take the effort to fix any serious vulnerabilities in the services they host. Conversely, operators with a low level of security consciousness will not take the effort to fix vulnerabilities or not monitor the security of their services at all.

As mentioned in section 3.3, this premise leads to the simple statement that the number of vulnerabilities listed by Shodan for an IP is directly influenced by the security consciousness of the operator of the IP. The security consciousness of IP operators can thus be measured indirectly by analysing the vulnerabilities.

The presence of one vulnerability is not independent of others: running an older, insecure version of a protocol often results in multiple vulnerabilities, and hosting many services increases the likelihood of vulnerabilities being present. Therefore, simply using the number of vulnerabilities on each host is a misleading and inaccurate metric. To this end, this study introduces a more sophisticated metric that is not influenced by the number of services or interdependencies between vulnerabilities.

To preserve accuracy, the metric will provide each host with a binary label based on one or more of three selected common vulnerabilities being present in the Shodan data. These vulnerabilities are specifically selected such that each meets the following criteria:

- The vulnerability concerns a service that is very common for internet hosting.
- The vulnerability causes a serious security risk.
- The vulnerability has been publicly known for at least two years.
- A public patch has been released not long after the vulnerability became public.
- The vulnerability is actively being detected by Shodan.

These criteria guarantee that any operator with high security consciousness will have known about the vulnerability for a long time and since taken the effort to fix it. Thus, any one of these kind of vulnerabilities being present is a strong indication that the IP operator has low security consciousness. Based on the criteria, the following vulnerabilities are selected:

- **CVE-2023-48795:** A vulnerability present in OpenSSH before version 9.6 which allows remote attackers to bypass integrity checks. This results in some security features for client-server connections being downgraded or even disabled [17].
- **CVE-2021-23017:** A problem in Nginx where an attacker who is able to forge UDP packets from the DNS server to cause 1-byte memory overwrite. Such an exploit can crash the Nginx server or even cause more severe remote memory control issues [15].
- **CVE-2023-25690:** A proxy configuration problem in the Apache HTTP Server allows for an HTTP request smuggling attack. This can result in bypass of access controls in the proxy server, proxying unintended URLs to existing origin servers, and cache poisoning [16].

4.3 ISP comparison

To answer the research question, this research compares the level of security consciousness of internet service hosts in the networks of different ISPs in the Netherlands. Using the aforementioned security consciousness metric, each host can be labelled as being conscious about security or not. The vulnerabilities in the metric are for the OpenSSH, Nginx and Apache HTTP services. These are among the most used services hosted by IPs from Dutch ISPs according to Shodan. However, there is still a difference in the fraction of IPs that host these exact services between the ISPs. Therefore, to prevent any bias being introduced by this difference, only the IPs that host at least one of these services are taken into account when checking if they have a vulnerability. These IPs are referred to as ‘service hosts’.

The ISPs are compared by computing the fraction of sampled service hosts which are labelled as having low security consciousness, i.e. the fraction of hosts that have at least one of the three vulnerabilities. To determine if this difference is actually significant and not due to variance in the samples, Fisher’s Exact Test is applied to each pair of ISPs with the following hypotheses [22]:

- **Null Hypothesis (H_0):** The two samples of service hosts come from the same distribution of low security conscious hosts.
- **Alternative Hypothesis (H_1):** The two samples of service hosts come from the different distributions of low security conscious hosts.

Since the test is two-sided, the significance threshold α is set to 0.025.

In the Netherlands, there are likely a couple dozen ISPs currently active. For this research, five are compared. These firstly include the three mainstream providers (KPN, Ziggo, Odido) which mainly aim to attract consumers with low prices, discounts and rewards [12, 18, 26]. These are contrasted against Freedom Internet, a smaller provider which has made security and privacy its main marketing points [10]. Lastly, the comparison includes Delta Fiber, a medium-sized provider which presents itself as being innovative and premium quality, but does not explicitly list security as one of its main qualities [5].

5 Results

Using Shodan, five samples of hosts are obtained for the five investigated ISPs. The combined sample size is 51749. Firstly, hosts that are obviously honeypots are identified and removed. The number of obvious honeypots in the samples is very small, with a total of 85. Then, all hosts which are not 'service hosts', i.e. which are not running OpenSSH, Nginx or Apache HTTP Server, are removed. This shrinks some of the samples by quite a lot: The KPN sample shrinks from almost 18000 to around 7500 and for the Freedom sample only 13% of hosts are service hosts, while for Delta the entire sample consists of service hosts. This difference is explained by the differences in the way each sample is obtained. Freedom is a small ISP, so every IP address it owns and is in the Shodan data is sampled. The complete IP range for Freedom is obtained using data from IPInfo. For Delta, which is also a smaller provider, Shodan filters are used to directly obtain all service hosts. Each of the other three samples is a mix of smaller samples of randomly obtained hosts and of those obtained while filtering specifically for service hosts. The exact sample sizes are presented in table 1.

ISP	Hosts	Honeypots	Real Hosts	Service Hosts
KPN	18016	27	17989	7574
Ziggo	10357	7	10350	7823
Odido	8024	46	7978	6562
Delta	4441	4	4437	4437
Freedom	10911	1	10910	1410

Table 1: Sample Sizes, obvious honeypots and service hosts per ISP.

Next, each service host is labelled as having low security consciousness or not using the metric described in section 4.2. Table 2 shows how many service hosts per ISP have which vulnerability and how many have at least one. The latter being the number of hosts labelled as having low security consciousness.

ISP	Hosts w/ SSH CVE	Hosts w/ Nginx CVE	Hosts w/ Apache CVE	Hosts w/ one or more
KPN	415 (5.48%)	175 (2.31%)	228 (3.01%)	795 (10.50%)
Ziggo	562 (7.18%)	208 (2.66%)	258 (3.30%)	996 (12.73%)
Odido	301 (4.59%)	171 (2.61%)	161 (2.45%)	662 (9.48%)
Delta	248 (5.59%)	104 (2.34%)	174 (3.92%)	513 (11.56%)
Freedom	75 (5.32%)	66 (4.68%)	79 (5.60%)	213 (15.11%)

Table 2: Number of service hosts with vulnerabilities. The rightmost column is the number of hosts labelled with low security consciousness.

Finally, the number of hosts with low security consciousness is compared between the groups. From the table above, one can see that KPN, Odido and Delta all have a fraction of hosts with low security consciousness close to 10%. Ziggo scores a bit higher with 12.7%. However, Freedom clearly scores higher than all the others with over 15%.

ISP	KPN	Ziggo	Odido	Delta	Freedom
KPN		0.000	0.046	0.073	0.000
Ziggo	0.000		0.000	0.059	0.016
Odido	0.046	0.000		0.000	0.000
Delta	0.073	0.059	0.000		0.001
Freedom	0.000	0.016	0.000	0.001	

Table 3: p-value according to Fisher's Exact Test for all combinations of ISPs. p-values indicating a significant difference between the groups ($p < \alpha$) are highlighted in green, and insignificant differences in red.

Table 3 shows the results of applying Fisher's Exact Test for all the combinations of ISPs. This confirms that most of the observed differences in security consciousness between the ISPs are significant. The table shows that Freedom is significantly different from all the other ISPs, with all p-values being between 0.000 and 0.016, well below the threshold of $\alpha = 0.025$.

5.1 Discussion

The results show that Freedom has the highest fraction of hosts with one or more selected vulnerabilities. This finding contradicts the hypothesis that Freedom's customers would demonstrate stronger security practices. Assuming that the data collection and statistical analysis are correct, several explanations can be considered from an actor perspective.

First, operators might not choose their ISP based on their own level of security consciousness. For most customers, factors such as price and connection speed are more decisive than perceived cybersecurity quality. A recent literature review on ISP selection criteria confirms this trend, decision-making is primarily driven by service quality, reliability, and performance rather than security considerations [14]. As a result, even if Freedom promotes itself as a security oriented provider, this does not translate into attracting a customer base that actively manages technical security on their own devices.

Secondly, it is possible that Freedom attracts a relatively high number of amateur tech-enthusiasts who do value privacy and security, but lack time or expertise to continuously maintain secure configurations. Literature frames this as a privacy/security paradox, where knowledge or motivation does not always translate into secure behaviour [19]. In contrast, professional or enterprise users who are more capable of mitigating vulnerabilities could be more present in larger ISP such as KPN or Ziggo. It is likely that a higher fraction of IPs from these ISPs is owned by large companies, who employ IT professionals to actively identify and fix vulnerabilities.

Thirdly, a behavioural effect called risk compensation could be a possible explanation, a phenomenon where individuals adjust their behaviour based on perceived safety [21]. Freedom's strong security focused branding may give customers a false sense of protection. As a result, users might neglect their own systems' security. This perception gap could lead to more unpatched or misconfigured services being exposed.

Exploring these literature-based possible explanations help to interpret the contradiction between the initial hypothesis and the results.

6 Limitations

6.1 Data limitations

The primary limitation of this research lies in the reliability of Shodan data. Shodan's vulnerability tags and banner information are not always entirely accurate or up to date, as shown in prior studies such as by Huang et al. and other evaluations of internet-wide scanners [11]. Banners can be missing, outdated, or intentionally modified by servers, and Shodan's automated mapping between software versions and CVEs is heuristic. Therefore, the results reflect *banner-visible indicators* of vulnerabilities rather than confirmed insecure systems. This uncertainty means that some hosts may appear vulnerable even after being patched, while others remain undetected despite running outdated software [6].

6.2 Methodology limitations

The selection of vulnerabilities also introduces methodological limitations. Only a specific set of CVEs was considered for being common, fixable, and visible in banners. This excludes many recent or less banner-detectable vulnerabilities, potentially biasing the analysis toward older or better-known issues. As prior research highlights, newer vulnerabilities are often under-represented in scanning datasets until banner fingerprints are updated [4]. In addition, differences in exploitability, patch timing, and software configurations mean that the presence of a CVE tag does not always correspond to equal levels of real-world risk across hosts.

Another limitation stems from counting each IP as vulnerable if it exposes *at least one* affected service. This approach does not differentiate between a host running a single outdated service and one running many, nor does it account for the overall service mix per ISP. Similar concerns are noted in large-scale network measurement literature [6], where self-hosted or multi-service environments naturally inflate exposure counts. As a result, ISPs with more self-hosting users or greater service diversity naturally appear to have more vulnerable IPs, even if their relative patching behaviour is similar to others.

Finally, the dataset may still include honeypots and research systems that intentionally mimic vulnerable services. Although filtering heuristics were applied, it is impossible to remove all honeypots reliably from Shodan data. The presence of many undetected honeypots could be a viable explanation for the results: Freedom truly being a very secure ISP who has deployed many more sophisticated honeypots than other ISPs explains why many hosts with vulnerabilities are found in their network. This limitation has also been discussed in related measurement work [1], where honeypots often distort exposure statistics despite attempts at exclusion.

7 Conclusion

This study investigated whether different levels of security consciousness among server operators correlate with their selection of Internet Service Providers in the Netherlands, addressing whether market mechanisms can organically improve internet security or whether regulatory intervention is necessary.

Using Shodan measurements for five major ISPs, the analysis found the opposite of what the hypothesis predicts: Freedom Internet, the provider that markets security and privacy most prominently, had the highest share of vulnerable hosts (15.11%), exceeding KPN (10.50%), Odido (9.48%), Delta (11.56%), and Ziggo (12.73%). Fisher's exact test indicates these differences are statistically significant ($p < 0.025$), leading to rejection of the hypothesised association between operator security consciousness and ISP selection.

The evidence suggests that demand-side market mechanisms do not naturally enhance internet security through customer ISP selection. Security-focused branding does not align with reduced vulnerability exposure among customers, and vulnerabilities do not cluster by providers' stated focus. Consistent with prior work showing that customers rely on indirect cues rather than measurable security outcomes, market signals alone appear insufficient to distinguish provider quality and reward better security practice. This indicates market failure in cybersecurity governance: customers cannot effectively distinguish between providers based on actual security outcomes, and security-focused marketing does not correlate with reduced vulnerability exposure.

The implications for multi-stakeholder governance are significant. If market differentiation cannot drive security improvements through informed customer choice, regulatory intervention may be necessary to establish minimum security standards and clarify responsibilities between ISPs and server operators. Future research should examine demographic differences between ISP customer bases, assess whether ISP-provided security services could bridge the gap between marketing and measurable outcomes, and develop longitudinal analyses to trace causal pathways across different regulatory contexts.

References

- [1] Michael Bailey, Evan Cooke, Farnam Jahanian, Jose Nazario, and David Watson. 2005. The Internet Motion Sensor: A Distributed Blackhole Monitoring System. In *Proceedings of the Network and Distributed System Security Symposium (NDSS)*. <https://www.ndss-symposium.org/ndss2005/internet-motion-sensor-distributed-blackhole-monitoring-system/>
- [2] Roland Bodenheimer, Jonathan Butts, Stephen Dunlap, and Barry Mullins. 2014. Evaluation of the ability of the Shodan search engine to identify Internet-facing industrial control devices. *International Journal of Critical Infrastructure Protection* 7 (2014), 114–123. Issue 2. doi:10.1016/j.ijcip.2014.03.001
- [3] Censys. 2021. *Moving Beyond the Noise by Filtering Internet Pseudo Services*. Retrieved 2025-10-09 from <https://censys.com/blog/beyond-noise-by-filtering-pseudo-services>
- [4] Roland Croft, M. Ali Babar, and Mehdi Kholoosi. 2023. Data Quality for Software Vulnerability Datasets. In *2023 IEEE/ACM 45th International Conference on Software Engineering (ICSE)*. 121–133. doi:10.1109/ICSE48619.2023.00022
- [5] Delta Fiber. 2025. *deltafiber.nl, homepage*. Retrieved 2025-10-09 from <https://www.deltafiber.nl/>
- [6] Zakir Durumeric, Michael Bailey, and J. Alex Halderman. 2015. A Search Engine Backed by Internet-Wide Scanning. In *Proceedings of the 22nd ACM Conference on Computer and Communications Security (CCS)*. 542 – 553. doi:10.1145/2810103.2813703
- [7] Zakir Durumeric, Hudson Clark, Jeff Cody, Elliot Cubit, Matt Ellison, Liz Izhikevich, and Ariana Mirian. 2025. Censys: A Map of Internet Hosts and Services. In *Proceedings of the ACM SIGCOMM 2025 Conference*. 147–163. doi:10.1145/3718958.3754344
- [8] A. Edu and M. Papadaki. 2011. Vulnerability Awareness. In *Advances in Networks, Computing and Communications 6*. Centre for Information Security and Network Research, University of Plymouth, 32–39. <https://books.google.nl/books?hl=nl&lr=&id=J6tGawAAQBAJ&oi=fnd&pg=PA32>
- [9] Federal Bureau of Investigation (FBI), Cybersecurity and Infrastructure Security Agency (CISA), Environmental Protection Agency (EPA), and Multi-State Information Sharing and Analysis Center (MS-ISAC). 2021. *Joint Cybersecurity Advisory: Compromise of U.S. Water Treatment Facility (AA21-042A)*. Technical

- Report. Federal Bureau of Investigation, Cybersecurity and Infrastructure Security Agency, Environmental Protection Agency, and Multi-State Information Sharing and Analysis Center. <https://www.cisa.gov/news-events/cybersecurity-advisories/aa21-042a>
- [10] Freedom Internet. 2025. *freedom.nl, homepage*. Retrieved 2025-10-09 from <https://www.freedom.nl/>
- [11] Szu-Chun Huang, Harm Griffioen, Max van der Horst, Georgios Smaragdakis, Michel van Eeten, and Yuri Zhauniarovich. 2025. Trust but Verify: An Assessment of Vulnerability Tagging Services. In *Proceedings of the 34th USENIX Security Symposium*. USENIX Association, 823 – 842. <https://dl.acm.org/doi/10.5555/3766078.3766121>
- [12] KPN. 2025. *KPN.com, homepage*. Retrieved 2025-10-09 from <https://www.kpn.com/>
- [13] Roberto J. Mejias, Joshua J. Greer, Gabrila C. Greer, Morgan M. Shepherd, and Raul Y. Reyes. 2025. A model for information security vulnerability awareness. *Computers & Security* 151 (2025), 104305. doi:10.1016/j.cose.2024.104305
- [14] Mohammed Naji, Sivadas Thiruchelvam, and Mohammed Khudari. 2023. An investigation of Internet Service Provider selection criteria: A systematic literature review. *Iraqi Journal for Computer Science and Mathematics* 4, 4 (2023). doi:10.52866/ijcsm.2023.04.04.013
- [15] National Institute for Standards and Technology. 2021. *National Vulnerability Database, CVE-2021-23017 Detail*. Retrieved 2025-10-09 from <https://nvd.nist.gov/vuln/detail/CVE-2021-23017>
- [16] National Institute for Standards and Technology. 2023. *National Vulnerability Database, CVE-2023-25690 Detail*. Retrieved 2025-10-09 from <https://nvd.nist.gov/vuln/detail/CVE-2023-25690>
- [17] National Institute for Standards and Technology. 2023. *National Vulnerability Database, CVE-2023-48795 Detail*. Retrieved 2025-10-09 from <https://nvd.nist.gov/vuln/detail/CVE-2023-48795>
- [18] Odido. 2025. *odido.nl, homepage*. Retrieved 2025-10-09 from <https://www.odido.nl/>
- [19] Athanasios Paspatis, Vasilis Katos, and Spiros Angelopoulos. 2023. Revisiting the Privacy Paradox in the Context of Security Awareness: Attitudes versus Behaviours. 7, 8 (2023). doi:10.3390/mti7080076
- [20] Soumya Ray, Terence Ow, and Sung S. Kim. 2011. Security Assurance: How Online Service Providers Can Influence Security Control Perceptions and Gain Trust. *Decision Sciences* 42, 2 (2011), 391–412. doi:10.1111/j.1540-5915.2011.00316.x
- [21] Roozmehr Safi and Glenn J. Browne. 2022. Detecting Cybersecurity Threats: The Role of the Recency and Risk Compensating Effects. *Information Systems Frontiers* 25 (2022), 1277–1292. doi:10.1007/s10796-022-10274-5
- [22] David J Sheskin. 2000. *Handbook of parametric and nonparametric statistical procedures, second edition*. Chapman and Hall/CRC.
- [23] Shodan. [n.d.]. *Understanding Shodan Vulnerability Assessment*. Retrieved 2025-10-01 from <https://help.shodan.io/mastery/vulnerability-assessment>
- [24] Andrea Tundis, Eric Marc Modo Nga, and Max Mühlhäuser. 2021. An exploratory analysis on the impact of Shodan scanning tool on the network attacks. In *ARES '21: Proceedings of the 16th International Conference on Availability, Reliability and Security*. 1–10. doi:10.1145/3465481.3469197
- [25] Shuaibu Hassan Usman. 2013. A Review of Responsibilities of Internet Service Providers Toward Their Customers' Network Security. *Journal of Theoretical and Applied Information Technology* 49, 1 (2013), 70–78. <https://www.jatit.org/volumes/Vol49No1/9Vol49No1.pdf>
- [26] Ziggo. 2025. *Ziggo.nl, homepage*. Retrieved 2025-10-09 from <https://www.ziggo.nl/>